

Premier University, Department of CSE
Spring 2026 , 4th Semester, Assignment
Course Title: Database Management Systems, **Course Code:** CSE 2221
Course Outcome: CO3, **Total Marks:** 10

An airline operates multiple flights between different airports and needs an efficient database system to manage its daily operations. The system must handle information related to airlines, airports, aircraft, flights, employees, passengers, bookings, and payments.

Objectives:

- Analyze the requirements of different stakeholders and convert them into a comprehensive database design.
- Understand and apply relational database concepts.
- Normalize a database design for efficiency and integrity.

Investigation:

Design a relational database for an Airline Flight Booking System that includes at least the following core entities: Airline, Airport, Aircraft, Flight, Employee (with specialization: Pilot, Cabin Crew, Ground Staff), Passenger, Booking, and Payment.

Evaluation:

- Discuss the reasons behind utilizing a relational database for this scenario.
- Design an EER (Entity-Relationship) diagram showing the relationships between the entities.
- Normalize the database design up to the third normal form (3NF).
- Clear and structured documentation.

Design:

Propose a comprehensive database design for the Airline Flight Booking System.

Deliverables:

- A printed assignment reporting the following tasks:
- A perfect database design for the Airline Flight Booking System addressing the objectives.
- Explanation of design choices and normalization process.
- Briefly address the complex problem-solving questions:
 - Does the solution need in-depth engineering knowledge?
 - Does the solution involve wide-ranging or conflicting technical, engineering, and other issues?

- Is the solution well-known, or does it require abstract thinking and analysis to formulate?
- Does the solution involve infrequently encountered issues?
- Does the solution need adherence to standards and codes of practice?
- Does the solution involve stakeholders with conflicting technical requirements?
- Does the solution involve interdependence between sub-problems or parts?

Rubrics for Assignment marking:

Task	Criteria	Good (4-5)	Moderate (2-3)	Poor (1)
i.	Problem solution	Properly or near appropriately reasoned solution	Appropriate solution for some cases	Inappropriate or no solution
ii.	Problem analysis	In-depth analysis	Shallow analysis	Incomplete analysis